Exercise 9.2

For $\mathcal{Y} = \mathbb{R}$, consider K-nearest-neighbor regression,

$$\hat{f}(x; D) = \frac{1}{K} \sum_{j \in N(x,D)} y_j,$$

where $N(x, D)$ indexes the K nearest feature vectors to x in D. Show that this algorithm has the affine dependence on the candidate response required by (9.4), so that its full conformal prediction set can be computed efficiently.

Solution Fix the training data $D_n = ((X_j, Y_j))_{j=1}^n$ and the test feature X_{n+1}. For a candidate response y, let

$$D_{n+1}^y = ((X_1, Y_1), \ldots, (X_n, Y_n), (X_{n+1}, y)).$$

Use a fixed rule for distance ties that depends only on features and indices, rather than response values. Then the neighbor set $N(x', D_{n+1}^y)$ does not depend on y: the augmented feature vectors are the same for every candidate. Denote this set by $N_{x'}$. Separating out the possible contribution of the test point gives

$$\begin{aligned} \hat{f}(x'; D_{n+1}^y) &= \frac{1}{K} \sum_{j \in N_{x'} \cap \{1,\ldots,n\}} Y_j + \frac{\mathbf{1}\{n+1 \in N_{x'}\}}{K}\, y \\ &= a(x'; D_n, X_{n+1}) + b(x'; D_n, X_{n+1})\, y, \end{aligned}$$

where

$$\boxed{a(x'; D_n, X_{n+1}) = \frac{1}{K} \sum_{j \in N_{x'} \cap \{1,\ldots,n\}} Y_j, \qquad b(x'; D_n, X_{n+1}) = \frac{\mathbf{1}\{n+1 \in N_{x'}\}}{K}.}$$

Both coefficients are independent of y, establishing (9.4).

To see explicitly why this helps, write $a_i = a(X_i; D_n, X_{n+1})$ and $b_i = b(X_i; D_n, X_{n+1})$. For the absolute residual score, the training and test scores are

$$S_i^y = |Y_i - a_i - b_i y| \quad (i \leqslant n), \qquad S_{n+1}^y = |(1 - b_{n+1})y - a_{n+1}|.$$

Thus

$$\mathcal{C}(X_{n+1}) = \left\{ y \in \mathbb{R} : \frac{1 + \sum_{i=1}^n \mathbf{1}\{|Y_i - a_i - b_i y| \geqslant |(1 - b_{n+1})y - a_{n+1}|\}}{n+1} > \alpha \right\}.$$

Each comparison can change only at a solution of one of the two linear equations

$$Y_i - a_i - b_i y = \pm\big((1 - b_{n+1})y - a_{n+1}\big).$$

Equations that hold identically contribute no boundary; each remaining equation has at most one solution. After computing the neighbor sets and coefficients once, there are therefore at most $2n$ candidate boundary points. Evaluate the p-value between consecutive boundary points and at the boundary points themselves to recover the exact prediction set. ////